

Can the v2.1 Sentiment Model Trade Stocks? — Consolidated Report

Author’s note. Consolidated write-up of RESULTS.md and the two external reviews. Study designed and directed by Russell Ding; analysis executed with Claude Code; independent reviews by Codex and Kimi. “You” in the text refers to the author. All conclusions verified by the author.

Correction (2026-09-03). The scores analysed here were produced by the **v2.0** model checkpoint, not v2.1 as the title and text say (the inference run predates the v2.1 release, the launcher notebook hard-codes v2.0, and re-scoring stored articles reproduces the saved values with v2.0 only). Read “v2.1” as “v2.0” throughout; no number changes. Details: RESULTS.md (correction note) and docs/project_report.pdf.

Three-agent analysis: Claude (study author) · Codex/GPT (reviewer) · Kimi (reviewer + replication) *Compiled 2026-08-09 · Source docs: signal_strategy/RESULTS.md, signal_strategy/notes/reviews/*

1. Executive summary

Question. The v2.1 entity-sentiment model reads financial news and scores how positive or negative each article is for each company mentioned. Can those scores predict stock returns and drive a trading strategy?

Answer, agreed by all three agents: No. The model **explains** stock moves — it does not **predict** them. The strong early results turned out to be a subtle form of look-ahead bias: about 25% of articles (including the 4–6 pm earnings spike) were dated to a trading day whose 4 pm close had *already happened* before the article was published. The market had already reacted; the “prediction” was really the model agreeing with a move that had already occurred. Once the data is restricted to news genuinely knowable before the close, predictive power is zero at every horizon, and every executable trading variant loses money.

What survives — also agreed by all three agents:

1. **The model is an accurate measurement instrument.** Its scores line up strongly with how the market actually reacts to news. That is valuable for monitoring, attribution, and analysis — just not for forecasting returns.
2. **One real predictive result: volatility.** The *magnitude* of sentiment (ignoring direction) predicts how much a stock will move the next day, even after controlling for recent volatility. Kimi independently verified this with an extra control and it held up.

The pivot both reviewers endorse: use the model for risk/volatility overlays, news monitoring, overnight P&L attribution, and a model-QA feedback loop — not for directional trading.

2. The data and the strategy design

Ingredients

Component	Detail
News	118,033
+	unique
sentiment	tier-1
	arti-
	cles,
	scored
	by
	the
	v2.1
	model
	into
	per-
	ticker
	sentiment
Universes	502
	tick-
	ers
	(cur-
	rent
	S&P
	500
	members)
Period	2020-
	03-23
	to
	2025-
	12-31
	(1,441
	trad-
	ing
	days)
Prices	EODHD
	daily
	OHLC
	+ ad-
	justed
	close
Panel	60,888
	ticker-
	days;
	me-
	dian
	45
	names
	per
	day
	with
	news

Signal construction

Each article’s per-ticker sentiment scores were combined into one number per ticker per day: **sent_mw**, a mention-weighted average (an article that talks about Apple ten times counts more for Apple than one that mentions it once). Multiple articles on the same ticker-day are averaged.

The strategy tested

The classic cross-sectional recipe: every day, **rank** all tickers with news by sentiment, **buy the top 20%** (most positive news) and **short the bottom 20%** (most negative), dollar-neutral, hold for H days, and measure whether the long side beats the short side. Two execution assumptions were compared:

- **“Paper”** — enter at the close of the signal day itself (turns out to be infeasible; this is where the look-ahead lived).
 - **“Tradable”** — enter at the next morning’s open (realistic).
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3. The analysis pipeline (what was actually run)

Five steps, each a standard technique, run in sequence (`signal_strategy/src/run_all.py`):

Step 1 — Build the panel (`build_panel.py`). Join news sentiment to prices; compute forward returns at 1, 5, and 20 days.

Step 2 — Information Coefficient (IC) (`ic_analysis.py`). Each day, take the rank correlation between today’s sentiment and the *next* N days’ return, then average across all days. An IC of zero means the signal knows nothing; even a good daily equity signal is typically only 0.02–0.05. The t-statistic (NW t , adjusted for autocorrelation) says whether the average is statistically distinguishable from zero (roughly, $|t| > 2$ is significant).

Step 3 — Fama-MacBeth regression (`fama_macbeth.py`). Same question but as a daily regression with control variables (size proxy, 1-month and 12-month momentum), to check the sentiment effect isn’t just momentum or size in disguise. The slope is “extra bps of next-day return per 1 standard deviation of sentiment.”

Step 4 — Event study (`event_study.py`). Line up all the extreme-sentiment days at $t=0$ and trace the average cumulative abnormal return for the next 20 days, top quintile vs bottom quintile. Shows the *shape* of the reaction and how long to hold.

Step 5 — Long-short backtest (`backtest_ls.py`). The actual strategy: daily quintile portfolios, both execution modes, transaction costs at 5 and 10 bps per side, turnover accounting, out-of-sample split (last 30% of dates).

Then two follow-ups triggered by your question about MOC execution:

Follow-up A — MOC feasibility (`moc_feasibility.py`). Audited every article’s actual publication timestamp (converted to New York time) against a 15:45 ET market-on-close order cutoff, rebuilt the signal using only genuinely knowable news, and re-ran the backtest.

Follow-up B — Feasible IC rerun. Rebuilt the statistical tests on the corrected panel (`panel_feasible.parquet`) to see whether the “predictive content” itself survived. It did not (Section 5).

4. First-pass results (later shown to be contaminated)

These are the numbers as first computed. They looked like a clean success story — which is exactly why they’re worth seeing before the correction.

IC (Step 2):

Horizon	Mean IC	NW t	% days positive
1 day	+0.0204	3.65	54.7%
5 days	+0.0166	2.92	53.8%
20 days	+0.0096	1.48	53.7%

Fama-MacBeth (Step 3): sentiment worth **+10.3 bps per standard deviation** for next-day returns ($t = 6.7$), surviving size and momentum controls at 1–5 days.

Event study (Step 4): top-vs-bottom quintile spread of **+35.6 bps by day 1** ($t = 10.1$), peaking at +39.9 bps around day 8, then flat. The positive-news side drifts up persistently; the negative-news side mean-reverts.

Backtest (Step 5): this is where the story cracked open:

H	Entry	Gross Sharpe	Gross bps/day
1	Signal-day close (“paper”)	+2.93	+25.4
1	Next open (“tradable”)	−0.72	−4.5
8	Signal-day close	+0.80	+2.5
8	Next open	−0.41	−1.2

The decomposition showed why: of the +36 bps day-one spread, **+40.9 bps is the overnight gap** (signal-day close to next open) and **−4.9 bps** is the tradable intraday part. The entire “edge” sat in a window you cannot trade with daily bars. First-pass verdict at this point: *“genuinely predictive but untradable.”*

That verdict was wrong, in an instructive way.

5. The discovery: it was look-ahead, not prediction

Your MOC question — “can we enter at the signal-day close and exit the next close?” — forced an audit of *when each article was actually published* versus which trading day it was assigned to. Finding:

~25% of articles were assigned to a trading day whose close happened *before* the article existed. The biggest chunk is the 4:00–6:00 pm earnings-release spike: an earnings report published Monday 5 pm was stamped `trade_date = Monday`, so the “prediction” measured Monday’s close against Tuesday’s — but by Monday’s close the article didn’t exist, and by Tuesday’s open the market had fully reacted to it.

So the model wasn’t forecasting the market. The model was reading Monday-evening news and the market was reading the same news Monday evening — and the panel construction made that agreement look like foresight.

Corrected results (signal restricted to news knowable by 15:45 ET on day t):

Horizon	Original IC (contaminated)	Feasible IC	Feasible NW t
1 day	+0.0204 (t 3.65)	−0.0067	−1.39
5 days	+0.0166 (t 2.92)	−0.0020	−0.39
20 days	+0.0096 (t 1.48)	+0.0025	+0.41

Zero predictive content at every horizon. And the executable strategies:

Variant (H=1, equal weight)	Gross Sharpe	Gross bps/day
Original signal, close entry (look-ahead)	+2.93	+25.4
Feasible signal, MOC close-to-close (implementable)	−0.55	−3.7
After-hours news, traded at next open	−0.18	−1.3

The feasible signal’s overnight gap is +1.5 bps (vs +40.9 contaminated) — news published during the trading day is already in the price by the close, and after-hours news is fully repriced by the next open. **The market absorbs this news faster than any daily-bar strategy can act, at every entry point.**

The one survivor: volatility. Sentiment *magnitude* |sent_mw| (feasible version) predicts next-day *absolute* returns with rank-IC +0.024 (NW t 5.3) after controlling for trailing 21-day realized volatility. Intense news tone today means a bigger move tomorrow — direction unknown. That is a usable non-directional signal (risk management, position sizing, options).

6. External review 1 — Codex (GPT)

Codex reviewed the study read-only (RESULTS.md + all source code) *before* the feasible-IC rerun existed, and independently reached the same central verdict:

“The headline claim of ‘real predictive content’ is not established ... [it] should be withdrawn” until the IC / Fama-MacBeth / event-study tests are rebuilt on timestamp-feasible panels.

That is exactly what Follow-up B then confirmed — the claim died when retested.

Codex’s issue list (ranked by its severity):

#	Severity	Issue	Plain-language meaning
1	Critical	IC/FM/Event-study all use the contaminated panel	“pre-diction” evidence shared the same look-ahead as the back-test — confirmed

#	Severity	Issue	Plain-language meaning
2	High	Event-study t=10.1 as-sumes independent events	Thousands of over-lapping same-day/same-ticker events counted as independent; the true t is far lower
3	High	“Size” control is same-day dollar volume—	It’s liquidity, not market cap— and same-day volume <i>re-acts to the news</i> , so it’s a contaminated control

#	Severity	Issue	Plain-language meaning
4	High	Not genuinely out-of-sample	The holding period H=8 was picked using the full sample; the 70/30 “OOS” split is a report, not a holdout
5	High	Survivorship + missing returns	Universe is <i>today’s</i> S&P 500 (losers excluded); missing price bars silently count as zero — worst in the short (bad-news) leg

#	Severity	Issue	Plain-language meaning
6	Med-high	Overnight adjusted de-composition mixes price bases	Adjusted close-to-close minus raw open-to-close, combined additively — not an exact identity around dividends/splits
7	Med-high	Calendar time assumptions unaided	Timestamp times-tamps as claimed UTC; AAPL price dates used as the NYSE calendar; half-days ignored

#	Severity	Issue	Plain-language meaning
8	Medium	Forced quin- tiles on tiny cross- sections	As few as 10 names → 2- stock legs; ties bro- ken by row order
9	Medium	Fixed HAC lag 5 for 20- day over- lap- ping returns	Statistical error bars too nar- row at long horizons
10	Medium	Cost model illus- tra- tive, not conservative	No spreads, bor- row, drift, or time im- pact — moot for a negative- gross strategy

#	Severity	Issue	Plain-language meaning
11	Medium	COVID Sample regime starts contamination the March-2020 low; no year-by-year break-out was reported	

Codex’s bottom line on scope: the honest claim is “**no detected edge,**” not “**edge is exactly zero**” — the feasible MOC Sharpe of -0.55 carries $t = -1.4$.

Codex’s suggested next tests (top of its 12-item list): rebuild every statistic on point-in-time panels at multiple cutoffs (9:30 / 15:45 / 16:00 / next open); bucket the original results by publication hour to isolate the contamination directly; placebo date-shifts (a contaminated signal peaks at the wrongly-assigned date); portfolio-level HAC inference instead of event-level t-stats; year-by-year breakouts with frozen rules.

7. External review 2 — Kimi

Kimi went a step further than reviewing: it **re-ran the numbers itself from the artifacts on disk**. Its verdict:

“The study’s updated conclusion is correct and the self-diagnosis is genuine, not cosmetic ... the original ‘predictive content’ was overwhelmingly look-ahead.”

Independent replications (the most valuable part):

- **A sharper contamination test.** Instead of rebuilding the panel, Kimi simply *dropped* the ~25% of ticker-days the feasibility rule reassigns and re-ran the IC on what remained: it collapsed from $+0.0204$ ($t\ 3.65$) to **$+0.0037$ ($t\ 0.68$)**. Over 80% of the headline IC came from the after-hours subset.
- **The volatility result survived a harder test.** Kimi added a control the study didn’t use — same-day absolute return — on top of trailing volatility. Result: partial rank-IC **$+0.019$ (NW $t\ 3.7$)**, stable in every year 2021–2025.
- **Sanity checks passed:** timezone/DST handling verified clean on ~60k timestamps; the contamination is *not* a COVID artifact (the original IC is positive every single year 2020–2025).

New problems Kimi found that Codex missed:

1. **The “late news at next open” test entered one session too late.** The late-news panel dated articles to the first knowable *close*, and the backtest then entered the *following* open — so Monday-evening news was traded at Wednesday’s open, not Tuesday’s. Kimi ran the correct Tuesday-open version itself: also 0 (-4.9 bps day-1 open-to-close, Sharpe -0.72). The conclusion stands, but the test as originally labeled was the wrong test.

2. **The decisive correction isn't reproducible.** `panel_feasible.parquet` and the feasible-IC/volatility tables were produced by ad-hoc uncommitted scripts. Kimi's recompute gives feasible 1-day IC -0.0006 ($t -0.10$) vs the reported -0.0067 ($t -1.39$) — same verdict (0), different exact numbers. And Fama-MacBeth + the event study were **never re-run** on the corrected panel, so those sections remain contaminated as printed.
3. **A hidden reversal hint.** The pooled feasible IC of 0 averages away two years — 2023 and 2024 — where it is significantly *negative* ($t -2$ each): possibly a short-horizon post-news reversal effect. Probably noise, but cheap to test and untested.
4. **One feasible trading channel was never tested:** pre-market news ($\sim 28\%$ of articles, published 5–9 am ET) traded at *that same day's* open. The current code can only enter the session *after* the signal day, so this — the last daily-bar stone unturned — was structurally impossible to express.
5. Smaller: the 15:45 rule ignores 1 pm half-day closes (~ 2 days/yr); $\sim 7k$ articles timestamped 0–5 am ET deserve a publication-vs-crawl-time audit.

8. Where the three agents agree and disagree

Full agreement (high confidence):

Point	Claude	Codex	Kimi
Directional	yes	yes	yes
daily	—	—	—
trad- ing: dead	tested	pre- dicted it	repli- cated it
Original	yes	yes	yes
“pre- dic- tion”	—	—	—
was look- ahead	found via MOC audit	flagged be- fore the rerun	quan- ti- fied: >80% of IC
Model is a good <i>mea- sure- ment</i> instrument	yes	yes	yes
Volatility	—	(post- dates its review)	yes — repli- cated, t 3.7 with extra control
sig- nal is real and the top sur- viv- ing use	t 5.3		

Point	Claude	Codex	Kimi
Event-study t=10 over-stated (non-independent events)	accepted	yes — flagged	yes — flagged
“Size” control is post-treatment dollar volume	accepted	yes — flagged	yes — flagged
Weak OOS / multiple- testing discipline	accepted	yes — flagged	yes — flagged (ex-tends it to the vol result)

Points of tension (worth knowing, none change the verdict):

- **Exact feasible-IC numbers.** Claude reported -0.0067 ($t = -1.39$); Kimi’s recompute from the same parquet gives -0.0006 ($t = -0.10$). Both are “zero”; the discrepancy exists because the correction was ad hoc, not scripted — Kimi’s reproducibility criticism, demonstrated on itself.
- **How dead is trading?** Codex and Kimi both stress the negative results are “no detected edge,” not proof of zero (t -stats around -1.4). And Kimi keeps one door ajar: the untested pre-market-news-at-open channel, plus the 2023–24 reversal hint. Weak priors on both, but they are honest open items rather than tested failures.
- **Half-corrected record.** Kimi is right that Fama-MacBeth and the event study were never re-run on the feasible panel — the correction so far rests on the IC and the backtest only.

9. Recommendations (merged from all three agents)

If you want to close out the trading question completely

1. **Commit a `feasible_analysis.py`** that rebuilds `panel_feasible.parquet` and re-runs IC + Fama-MacBeth + event study on it, so the correction is a runnable script, not a claim. (*Kimi’s #1; Codex’s #1.*)
2. **Test pre-market news at the same day’s open** — the one untested feasible channel (~28% of articles). (*Kimi.*)
3. **Check the 2023–24 negative-IC reversal hint** with a proper short-horizon reversal test. (*Kimi.*)

4. If any of those show life, apply the full methodology hardening: portfolio-level HAC inference, lagged controls, point-in-time universe, frozen rules with a one-shot holdout. (*Codex.*)

The productive pivot: non-trading uses (priority order)

1. **Volatility / risk overlay** — the surviving predictive result. Feed $|\text{sentiment}|$, dispersion, and news intensity into volatility forecasts, position sizing, VaR, or options strategies. Gate on hardening first: earnings-day control, overnight-vs-intraday split of the predicted move, a true out-of-sample freeze. (*All three agents' top pick.*)
2. **Overnight P&L attribution** — the model demonstrably explains overnight gaps; entity-level granularity lets you attribute a portfolio's overnight move to specific companies and articles. Turns the study's fatal flaw into the product. (*Kimi #2, Codex #4.*)
3. **News monitoring and alerts** — flag sentiment shocks and deterioration on held names; rank the daily news queue by relevance, extremity, and corroboration (multi-article days carry more signal — the study's own intensity split). Latency-insensitive when framed as a review queue.
4. **Model-QA flywheel** — the panel already holds model sentiment, vendor polarity, and realized market reaction side by side; disagreements between the three surface mislabeled or hard articles for the next retrain. Directly feeds the parent project. (*Codex #7-8, Kimi #4.*)
5. **Slow fundamental features** — weekly/monthly sentiment level, dispersion, and negative-tail frequency for slower models. Lower priority: the feasible IC is 0 even at 20 days, so pitch these as features, not signals.

Explicitly deprioritized

Any directional daily-bar strategy — close entry, next-open entry, MOC — all tested, all negative, and the mechanism is understood: **the market prices this news faster than daily bars can act.**

Appendix — mini-glossary

- **bps** — basis points; 1 bp = 0.01%.
- **IC (Information Coefficient)** — daily rank correlation between the signal and subsequent returns, averaged over days. 0 = no skill; 0.02–0.05 is typical for real (weak) equity signals.
- **NW t** — Newey-West t-statistic: significance adjusted for the fact that daily results overlap/correlate. Roughly, $|t| > 2$ = statistically significant.
- **Fama-MacBeth** — run one cross-sectional regression per day, then average the daily slopes; the standard way to test a return predictor with controls.
- **Sharpe ratio** — annualized return divided by annualized risk. > 1 is good; negative means it loses money.
- **MOC** — market-on-close order; must be submitted by ~15:45–15:50 ET, which is why 15:45 is the study's “knowability” cutoff.
- **Look-ahead bias** — using information in a backtest that was not actually available at the assumed decision time. The central villain of this study.
- **Overnight gap** — price change from one day's 4 pm close to the next morning's 9:30 open; where most earnings reactions happen, and untradable with daily-bar entries.

File map: study details RESULTS.md · code src/ · reviewer prompts and full raw reviews notes/reviews/{review_prompt.md, codex_review_raw.txt, kimi_review_raw.txt} · corrected panel outputs/panel_feasible.parquet · MOC study outputs/moc_feasibility.json.